



Financial Institutions & Markets- ENT 309

PPT #9, IITD, Summer, 2026

Agenda

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STOCK MARKETS

- Stock Market Structure
- A Few Concepts
- Valuation of Stocks
- A Few Examples

MUTUAL FUNDS

Stock Market Structure & Few Concepts



Market Participants

- Regulator: SEBI (Securities & Exchange Board of India)
- Depositories: Store and hold stocks in demat form
 - NSDL: National Securities Depository Ltd
 - CDSL: Central Depository Service Ltd
- Exchanges: BSE, NSE
- Brokers: Facilitate transactions at exchanges
- Clearing Corporations: Guarantee settlement of transactions. Take care of counterparty risk. NSE Clearing Ltd, Indian Clearing Corporation Ltd, Clearing Corporation of India Ltd.
- Retails & Institutional Buyers & Sellers and Traders



A few concepts

- What does Limited or Ltd. in the name of a company mean?
 - Limited Liability
 - Co. is a distinct entity that the owners/ shareholders
 - Their losses limited to the amount invested in the company
- A company has an Authorized Capital and a given number of stocks. This can be increased/ decreased by following a specific process.
- What does a company do with its earnings/ its profit?
 - **Pay dividend** to shareholders OR
 - **Add to Reserves & Surplus** – Retain for use in companies future investment and growth. Retained Earnings.



A few concepts

- If Retained Earnings contribute to growth, co. becomes more valuable and share price goes up.
- Primary Market
 - Cos. do an IPO (Initial Public Offer). Cost is involved.
 - Have to submit an application with details to SEBI for approval
 - Red Herring Prospectus (DRHP): Biz model, financials, growth plans
 - Issue underwritten by Investment Bankers. If public does not buy the issue fully, they buy the devolved portion. If <90% subscription, then issue is cancelled.
 - INFY: Infosys share was undersubscribed when its IPO was done.



A few concepts

- Secondary Market
 - Buy & Sell once the IPO has been completed
 - BSE & NSE
- Private Placement Of Shares: Company can offer shares to a select group of persons. And not make an open, public offer. Do IPO later.
- SME Exchange: At BSE & NSE. For Small & medium enterprise who are not ready to go to mainboard BSE/NSE.
- Intraday Trading: Buy & Sell the same day
- Interday : Across many days. Buy and hold
- Stop Loss: Specify a price. Stock automatically sold if that price is reached. A safety net.



A few concepts

- Margin Trading: Investor buys stocks by paying a fraction of the price (margin) and borrowing the remaining amount from the broker by paying interest.
 - Profit or Loss is magnified.
- Limit Order: Maximum price willing to pay when buying a stock. Minimum willing to accept when selling a stock. Order executed when condition met.
- Short Selling: If you expect the price of a stock to go down, you can sell the shares (without owning them yet) and buy them later in the day. A margin amount is kept with the broker to cover his risks. Only Intraday allowed for retail investors in India.



Types of Stocks

- Holding a stock gives you ownership of the company
- Common Stock
 - **Residual claim** on company's cash flows & assets. If co. goes bankrupt, then common stock holders will be paid after employees, government, bond holders etc. have been paid.
 - Dividends: Not guaranteed
 - Have voting rights
- Preferred Stock
 - Co. has a **fixed** obligation to pay dividend. Like a bond. Fixed Dividend.
 - Not paying dividend does not mean default.
 - Priority over Common Stock. Dividend Payment or liquidation of Assets
 - No voting rights
 - <https://www.screener.in/screens/421894/preferred-stocks/>

Exchange Traded Funds ETFs

- A stock created from a basket of stocks. Like a Mutual Fund (MF)
- Listed & traded like a stock on exchange
- Price based on the price of the constituent stocks
- Passively follow the basket and not actively managed.
- Can use limit order, short selling, stop loss, margin trading etc.
- Like in case of a stock, brokerage incurred in buying & selling.
- But lower expense ratio as compared to MF.
- <https://www.nseindia.com/market-data/exchange-traded-funds-etf>



Rewarding Shareholders

- Dividend Payout
 - Companies in growth industries retain more profits and grow in value.
 - Companies in mature industries with stable growth pay more dividends
- Bonus Issue: More stocks given in proportion to stocks already held
- Rights Issue: Given an option to buy more stocks at a price lower than market price

Valuation of Stocks



Research Approaches

- Top-Down Approach:
 - Macroeconomic Analysis
 - GDP growth, Inflation, Interest Rate cycle, Currency movement etc.
 - Sector & Industry
 - Demographic changes, technology changes, government policy, commodity price cycles
 - Specific company/share
 - Revenue growth, profit margins, governance & management, valuation multiples
- Bottom-Up Approach: Prioritize company analysis. A well run company can perform differently than the broader market & sector. Helps one find 'hidden gems'



Review: Few Concepts

- Price = Current (discounted) Value of all Future Cash Flows
- Holding Period Return = Return taking into account buying price, coupons earned, capital gain or loss on selling
- Rate of discounting = Risk Free Rate + Spread (based on risk assessment or risk rating of the company)
- Information Asymmetry & Regulations
 - Risk assessment depends on the information available. Which varies from one person to another
 - Insider Trading: Using non-public & material information to gain unfair advantage

<https://www.youtube.com/watch?v=ZgKEoNVVSCO>



Stock Valuation

- Dividend is equivalent to coupon payment in bonds
- Difference: Coupon payment fixed & known. Dividend payout variable & to be estimated- could be zero also.
- Retained earnings also increase value of company and stock
- Price of a stock responds to new information as it periodically becomes publicly available

Stock Valuation



Dividend Discount Model (DDM)

$$P_0 = \frac{D_1}{(1 + k_e)^1} + \frac{D_2}{(1 + k_e)^2} + \cdots + \frac{D_n}{(1 + k_e)^n} + \frac{P_n}{(1 + k_e)^n}$$

P_0 : Price today, D_i : Estimated Dividends, P_n : Selling price after n time periods, k_e : Discount Rate/ Required Return/ Opportunity Cost

- If P_n is far in the future, its discounted value will be very low.
- It can also be replaced with discounted future dividends
- Hence

$$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1 + k_e)^t}$$

Stock Valuation



Gordon Growth Model

- Assume
 - Dividends grow at a constant rate g . So $D_i = D_0(1 + g)^i$
 - $g < k_e$. Else the firm will grow impossibly large

$$P_0 = \frac{D_0 \times (1 + g)^1}{(1 + k_e)^1} + \frac{D_0 \times (1 + g)^2}{(1 + k_e)^2} + \dots + \frac{D_0 \times (1 + g)^\infty}{(1 + k_e)^\infty} \quad P_0 = \frac{D_0 \times (1 + g)}{(k_e - g)} = \frac{D_1}{(k_e - g)}$$

- CAN REFINE FURTHER
 - Initial Growth Phase: Assume a certain value of g - *the growth rate of dividends*
 - Later Mature Phase: After n years, assume a lower value of g
- What about the effect of retained earnings?
 - Captured in value of g . Retained earnings increase future growth rate of D



Stock Valuation

You are evaluating a stock expected to pay dividends of \$2 next year ($D_1 = \2). The firm is expected to grow at 3% infinitely ($g = 3\%$). You are quite uncertain about both the constancy of the dividend stream and the accuracy of the estimated growth rate. To compensate yourself for this risk, you require a return of 15%.

Jennifer, another investor, has spoken with industry insiders and feels more confident about the projected cash flows. Jennifer requires only a 12% return because her perceived risk is lower than yours.

Bud is a friend of the CEO of the company. He knows with near certainty what the future of the firm actually is. He thinks that both the estimated growth rate and the estimated cash flows are lower than what they will actually be in the future. Because he sees almost no risk in this investment, he requires only a 7% return.



Stock Valuation

- What will be the expected Stock Price by each individual?

Investor	Discount Rate	Stock Price
You	15%	\$16.67
Jennifer	12%	\$22.22
Bud	7%	\$50.00

- What will be the market price?
- Just above \$22.22.
- Bud is willing to pay any value below \$50. In this case, the next highest expectation is \$22.22. So at any price above that, Jennifer and you both will sell to Bud.

Stock Valuation



Price Earnings Valuation Method

- What if the firm is not paying dividends or erratic payouts?
- Price to Earnings (PE) is a popular ratio. Stock Price/Earning Per Stock
- Firms in the same industry expected to have same PE ratio
- If PE ratio of a company is high
 - Market expects Earnings to grow faster so that Future PE of Stock ~ PE of other firms in the sector
 - If future earnings growth does not meet expectations, the Price will fall- the stock might turn out to be overvalued
- If PE is low and there does not seem to be a problem, the stock might be undervalued

Stock Valuation



Price Earnings Valuation Method

The average industry PE ratio for companies similar to Jindal Steel is 18. The earnings per share next year are projected to be Rs 50. What would be the expected share price?

$$\begin{aligned}\text{Price} &= \text{PE Ratio} \times \text{Projected Earnings} \\ &= 18 \times 50 = \text{Rs } 900.\end{aligned}$$

Stock Valuation: Errors



- Errors in estimating growth of the firm
- Challenges in estimating risk
- Problems with forecasting dividends

Stock Market Indexes



- An index monitors the behaviour of a group of stocks. Mkt cap as weights
 - BSE Sensex: 30 largest companies
 - Nifty 50: NSE. 50 largest companies
 - Nifty 500: NSE. Top 500.
 - Nifty Midcap 100
 - Dow Jones Industrial Average: Top 30.
 - S&P 500 Index
 - NASDAQ composite index: All stocks on Nasdaq.
 - NYSE Composite: All stocks on NYSE.
- <https://groww.in/indices/indian-indices>

Mutual Funds



Mutual Fund Basics

- Regulator: SEBI (Securities & Exchange Board of India)
- Asset Management Co. (AMC)/ Mutual Fund House: Licensed Entities.
- Pool the money to buy stocks, bonds, money market instruments
- Each MF has a theme – investment focus
 - Equity Fund: Invest in stocks
 - Debt fund: Invest in bonds. Govt & Corporate bonds
 - Liquid fund: Invest in money market (short term bonds and overnight debt market)
 - Hybrid funds: Mix of equity & bonds.
- Hedge Funds: A special type of MF.

<https://www.youtube.com/watch?v=P3YPofukvnl>



Mutual Fund Basics

- Fund Managers: Experts who manage the fund and take decisions
- Charge a fund management fee- % of Assets Under Management (AUM)
- Actively Managed: Fund manager makes decisions to buy and sell
- Passively Managed: Index funds. MF tracks an index. Hence lower cost
- MF Units are allocated to investors.
- End of Day Net Asset Value (NAV) calculated. Trxns happen at NAV.
- Settlement happens after T+1 or T+2 depending on the fund type
- You buy from and sell to the fund house. No exchange involved.



MF Advantages

- Liquidity: Buy or sell at any time in any amount
 - Minimum amount: INR 100
 - Sometimes exit fee might be applicable
- Diversification: Investment spread across multiple stocks/bonds. Hence lowers the risk as compared to investing in one stock/bond.
- Fund Manager Expertise: The investor benefits from teams of analysts & experts
- Cost Averaging: By buying small amounts at regular intervals.
- Mutual Funds can pay dividends
- Can invest in international markets



MF Structure

- Closed Ended: Units sold initially. No exit for given number of years.
 - Limited number of funds. For investment themes which play out over a few years
 - Some closed ended MFs offer tax benefits
- Open Ended: Can buy & sell anytime
- Cost to investor
 - Entry load: Fees paid when buying a fund. Might be there
 - Exit load: Fees paid when selling a fund. Might be there
 - Expense Ratio: MF management fees and other charges ex. distribution charges



MF Structure

Same fund can be bought under (MF example)

- Regular Plan: Bought through advisor or broker. Distribution fees paid.
- Direct Plan: Transaction directly with fund house. Less costly as no distribution fees paid.
- Growth: Does not give dividends. Only capital gains
- Dividend: Pays regular dividends
- SIP (Systematic Investment Plan)- A fixed amount of money invested at regular interval. Automated. Daily, weekly, monthly etc.
- AUM: INR 82 trillion (May '26). INR 14 trillion (May 2016). 6 X in 10 yrs
- Explore <https://www.amfiindia.com>



Thank You!